

Model Lab

Every prior, same slate — 2026-09-12

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

34,754

Held-Out Scores

Fitted on plate appearances before 2026-08-08, scored on the 34,754 that came after (actual HR rate 2.93%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5684	1.33×	0.02839	0.13157	0.00475
KNN comps Prior is the pooled rate of the nearest comparables.	0.5905	1.57×	0.02837	0.13113	0.00458
SVR Support-vector regression on the same feature matrix.	0.5789	1.44×	0.02847	0.13301	0.00880
Random forest Forest regression; a learned similarity metric.	0.5810	1.42×	0.02846	0.13253	0.00853
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5905	1.57×	0.02837	0.13113	0.00458
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5900	1.54×	0.02837	0.13117	0.00478
Linear regression Ordinary least squares on the same features.	0.5858	1.40×	0.02843	0.13205	0.00725
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5876	1.50×	0.02844	0.13179	0.00771
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5874	1.46×	0.02841	0.13158	0.00719
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5815	1.43×	0.02852	0.13281	0.00924

Blend weights (forest variant): KNN 0.95, SVR 0.00, RF 0.05

Shrinkage K per prior: league 254, knn 254, svr 254, rf 254, core 254, ensemble 254, linear 254, logistic 254, nn 254, xgb 254 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,736 PA, actual HR rate 2.41%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5290	1.22×	0.02354	0.11416	0.00851
KNN comps Prior is the pooled rate of the nearest comparables.	0.5725	1.55×	0.02358	0.11538	0.01097
SVR Support-vector regression on the same feature matrix.	0.5491	1.11×	0.02409	0.12619	0.02546
Random forest Forest regression; a learned similarity metric.	0.5573	1.33×	0.02420	0.12398	0.02664
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5725	1.55×	0.02358	0.11538	0.01097
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5730	1.55×	0.02360	0.11563	0.01171
Linear regression Ordinary least squares on the same features.	0.5737	1.22×	0.02399	0.12237	0.02184
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5752	1.11×	0.02406	0.11962	0.02394
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5770	1.00×	0.02378	0.11778	0.01949
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5686	1.00×	0.02468	0.12611	0.03365

Slate Agreement

Flat league · 3/6 shared 1/6 hit

Pete Crow-Armstrong

Jackson Chourio

Eugenio Suárez ✓

Jeremy Peña

Joc Pederson

Alex Bregman

SVR · 3/6 shared 0/6 hit

Pete Crow-Armstrong

Jackson Chourio

Shea Langeliers

Junior Caminero

Aaron Judge

Oneil Cruz

Core (KNN+SVR) · 6/6 shared 0/6 hit

Pete Crow-Armstrong

Jackson Chourio

Cal Raleigh

Oneil Cruz

Joc Pederson

Jake Bauers

Linear regression · 4/6 shared 0/6 hit

Pete Crow-Armstrong

Aaron Judge

Cal Raleigh

Jackson Chourio

Shea Langeliers

Joc Pederson

Neural net (torch) · 4/6 shared 0/6 hit

KNN comps · 6/6 shared 0/6 hit

17.5% Pete Crow-Armstrong

17.1% Jackson Chourio

15.9% Cal Raleigh

15.8% Oneil Cruz

15.8% Joc Pederson

15.7% Jake Bauers

Random forest · 4/6 shared 0/6 hit

20.3% Pete Crow-Armstrong

19.2% Jackson Chourio

19.2% Jake Bauers

19.2% Aaron Judge

19.1% Joc Pederson

18.8% Shea Langeliers

Forest blend · 6/6 shared 0/6 hit

19.3% Pete Crow-Armstrong

18.4% Jackson Chourio

18.3% Cal Raleigh

18.2% Oneil Cruz

17.8% Joc Pederson

17.7% Jake Bauers

Logistic regression · 3/6 shared 0/6 hit

19.2% Pete Crow-Armstrong

19.1% Aaron Judge

18.8% Cal Raleigh

18.6% Shea Langeliers

17.7% Jackson Chourio

17.6% Kyle Schwarber

19.3%

18.4%

18.3%

18.2%

17.8%

17.7%

19.9%

19.5%

18.6%

18.5%

18.5%

18.2%

19.3%

18.4%

18.3%

18.1%

17.9%

17.8%

20.5%

20.4%

19.9%

19.4%

18.9%

18.3%

Jake Bauers

17.9%

Shea Langeliers

18.5%

Hitter by Hitter

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Shea Langeliers Athletics · vs LHP · 23 HR / 416 PA	14.0%	17.5%	19.2%	18.2%	17.5%	17.5%	17.7%	19.4%	18.7%
9. Eugenio Suárez Cincinnati Reds · vs RHP · 22 HR / 450 PA	15.9%	17.0%	17.6%	17.6%	17.0%	17.1%	17.2%	17.4%	17.2%
10. Juan Soto New York Mets · vs LHP · 24 HR / 412 PA	13.2%	16.9%	17.4%	17.3%	16.9%	16.9%	16.9%	17.1%	17.2%
11. Brandon Lowe Pittsburgh Pirates · vs RHP · 29 HR / 600 PA	14.8%	16.6%	16.9%	17.0%	16.6%	16.6%	16.6%	17.5%	17.0%
12. Corey Seager Texas Rangers · vs RHP · 17 HR / 375 PA	13.3%	16.4%	16.1%	16.6%	16.4%	16.4%	16.4%	16.6%	16.8%
13. Sal Stewart Cincinnati Reds · vs RHP · 31 HR / 636 PA	15.2%	16.3%	16.8%	17.0%	16.3%	16.3%	16.4%	16.1%	16.4%
14. Tyler Stephenson Cincinnati Reds · vs RHP · 15 HR / 423 PA	14.1%	16.2%	14.5%	15.3%	16.2%	16.1%	15.2%	14.8%	14.8%
15. Michael Conforto Chicago Cubs · vs RHP · 14 HR / 270 PA	13.5%	16.0%	15.5%	15.8%	16.0%	16.0%	15.4%	15.2%	15.6%
16. Jeremy Peña Houston Astros · vs RHP · 17 HR / 424 PA	15.8%	16.0%	16.4%	16.1%	16.0%	16.0%	15.7%	15.7%	15.7%
17. Junior Caminero Tampa Bay Rays · vs LHP · 40 HR / 643 PA	14.9%	16.0%	19.2%	16.8%	16.0%	16.0%	16.5%	17.1%	17.1%
18. Spencer Jones New York Yankees · vs RHP · 9 HR / 239 PA	13.9%	15.9%	15.9%	15.3%	15.9%	15.9%	14.5%	14.4%	14.9%
19. Kyle Schwarber slate Philadelphia Phillies · vs LHP · 44 HR / 637 PA	13.3%	15.8%	16.9%	16.4%	15.8%	15.8%	16.3%	18.3%	16.5%
20. Dominic Canzone Seattle Mariners · vs RHP · 21 HR / 470 PA	14.0%	15.7%	16.2%	16.7%	15.7%	15.8%	15.9%	16.3%	16.2%
21. Pete Alonso Baltimore Orioles · vs RHP · 34 HR / 642 PA	13.4%	15.7%	16.1%	15.7%	15.7%	15.7%	15.6%	16.2%	15.8%
22. Hunter Goodman	14.0%	15.6%	16.4%	16.1%	15.6%	15.6%	15.3%	16.0%	16.1%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
Colorado Rockies · vs RHP · 37 HR / 526 PA									
23. Matt Olson Atlanta Braves · vs RHP · 38 HR / 640 PA	13.5%	15.6%	16.7%	16.3%	15.6%	15.6%	15.7%	17.0%	16.5%
24. James Wood Washington Nationals · vs LHP · 30 HR / 570 PA	12.5%	15.6%	16.5%	15.8%	15.6%	15.6%	15.4%	16.0%	16.2%
25. Cam Smith Houston Astros · vs RHP · 20 HR / 517 PA	14.1%	15.5%	15.7%	15.5%	15.5%	15.5%	15.3%	15.4%	15.6%
26. Miguel Vargas Chicago White Sox · vs LHP · 30 HR / 639 PA	12.6%	15.5%	15.2%	14.8%	15.5%	15.5%	15.5%	16.0%	15.6%
27. Zack Gelof Athletics · vs LHP · 17 HR / 378 PA	13.7%	15.5%	15.5%	15.0%	15.5%	15.5%	15.5%	15.4%	15.5%
28. Elly De La Cruz Cincinnati Reds · vs RHP · 26 HR / 571 PA	14.4%	15.5%	16.5%	15.5%	15.5%	15.5%	15.3%	15.6%	15.4%
29. Alex Bregman Chicago Cubs · vs RHP · 25 HR / 661 PA	15.7%	15.5%	16.0%	15.7%	15.5%	15.5%	15.9%	15.9%	15.6%
30. Francisco Lindor New York Mets · vs LHP · 18 HR / 397 PA	14.5%	15.4%	15.4%	15.1%	15.4%	15.4%	15.3%	15.5%	15.1%
31. William Contreras Milwaukee Brewers · vs LHP · 16 HR / 592 PA	14.9%	15.3%	15.5%	15.0%	15.3%	15.2%	14.9%	15.0%	14.9%
32. Eduardo Valencia Detroit Tigers · vs LHP · 6 HR / 95 PA	12.5%	15.1%	15.1%	14.9%	15.1%	15.1%	15.6%	15.6%	15.9%
33. Kyle Stowers Miami Marlins · vs RHP · 19 HR / 463 PA	12.4%	15.1%	14.8%	14.6%	15.1%	15.0%	13.9%	14.9%	14.8%
34. Zach Neto Los Angeles Angels · vs RHP · 25 HR / 634 PA	13.1%	15.0%	15.1%	14.7%	15.0%	15.0%	14.3%	15.1%	14.4%
35. Coby Mayo Baltimore Orioles · vs RHP · 23 HR / 398 PA	12.5%	15.0%	16.0%	15.7%	15.0%	15.0%	15.4%	17.0%	15.9%
36. JJ Bleday Cincinnati Reds · vs RHP · 22 HR / 493 PA	13.0%	15.0%	15.4%	14.9%	15.0%	15.0%	14.6%	14.7%	14.7%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Griffin Conine Miami Marlins · vs RHP · 15 HR / 272 PA	12.1%	14.8%	15.7%	15.0%	14.8%	14.8%	14.8%	15.1%	15.3%
38. Max Muncy Los Angeles Dodgers · vs RHP · 27 HR / 508 PA	12.2%	14.7%	14.9%	14.6%	14.7%	14.7%	14.5%	15.5%	14.9%
39. Ronald Acuña Jr. Atlanta Braves · vs RHP · 16 HR / 417 PA	13.2%	14.6%	14.3%	14.1%	14.6%	14.5%	14.1%	14.2%	14.3%
40. Joshua Báez St. Louis Cardinals · vs RHP · 7 HR / 96 PA	11.4%	14.5%	15.4%	14.5%	14.5%	14.5%	14.0%	14.4%	15.2%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.